Algebraic K-theory
Lectures by Daniil Rudenko

Typed by Andrea Bourque

Spring 2026

Preface

There will be some gaps in explanation, either due to the lecturer's admission or my own lack of understanding. In particular, many "proofs" are sketches of proofs. Gaps due to my own misunderstanding will be indicated by three red question marks: ???. More generally, my own questions about the material will also be in red. Things like "**Question**" will be questions posed by the lecturer. Feel free to reach out to me with explanations.

Contents

1 Mar 23 **2**

2 Mar 25 **5**

2.1 End of overview 6

3 Mar 30 **7**

1 Mar 23

Objects of study:

- Algebraic K-groups, i.e. for ring R, groups $K_n(R)$ for $n \geq 0$.
- Steinberg module, a representation $St_n(F)$ of $GL_n(F)$ for F a field.

We assume throughout the course that rings are commutative. R will always denote a ring and F will always denote a field.

Definition 1.1. An R-module P is projective if there is an R-module P' such that $P \oplus P'$ is free.

Definition 1.2 (1957, Grothendieck)**.** The group $K_0(R)$ is generated by isomorphism classes of finitely generated projective R-modules, subject to the relation $[P \oplus Q] = [P] + [Q]$.

Example 1.1. If R is a PID (including a field), then $K_0(R) \cong \mathbb{Z}$, since every finitely generated projective R-module is free.

Example 1.2. If F is a number field with integer ring $\mathcal{O}_F$, then $K_0(\mathcal{O}_F) \cong \mathbb{Z} \oplus Cl(F)$, where $Cl(F)$ is the ideal class group.

Definition 1.3 (1964, Bass)**.** Consider embeddings $GL_n(R) \hookrightarrow GL_{n+1}(R)$ where A is sent to the block matrix $\begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}$. Let $GL(R) = \varinjlim GL_n(R)$. Then $K_1(R)$ is the abelianization of $GL(R)$, i.e. $GL(R)/[GL(R), GL(R)]$.

Another description of $[GL(R), GL(R)]$ is given as follows. For $i \neq j$ in $\{1, \ldots, n\}$, let $e_{ij}(a) = \mathrm{id} + a\delta_{ij}$ be the matrix in $GL_n(R)$ with 1 on the diagonal, a in the (i, j) entry, and 0 elsewhere. This is called an elementary matrix. Let $E_n(R)$ be the subgroup of $GL_n(R)$ generated by the elementary matrices. Then let $E(R) = \varinjlim E_n(R)$.

Theorem 1.1 (Whitehead)**.** $E(R) = [GL(R), GL(R)]$.

Thus $K_1(R) = GL(R)/E(R)$.

Example 1.3. If F is a field, then $K_1(F) \cong F^\times$. We also have $K_1(\mathbb{Z}) \cong \mathbb{Z}/2$.

Remark. If R is a regular Noetherian ring, then $K_i(R[t]) \cong K_i(R)$ for $i = 0, 1$.

Elementary matrices have certain "universal" relations that do not depend on the ring R, such as $e_{ij}(a)e_{ij}(b) = e_{ij}(a+b)$. This can be depicted with a picture:

$$\begin{tikzcd} & \bullet \arrow[dr, "e_{ij}(b)"] & \\ \bullet \arrow[ur, "e_{ij}(a)"] \arrow[rr, "e_{ij}(a+b)"'] & & \bullet \end{tikzcd}$$

If $i \neq l$ and $j \neq k$, then $[e_{ij}(a), e_{kl}(b)] = 0$ possibly id_n, I need to check. This is depicted by a square. If $i \neq k$, then $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$, which is depicted

by a pentagon. Note: lecturer did not label the edges of the square and pentagon, so I will not include their pictures.

The Steinberg group (unrelated to Steinberg module) $St_n(R)$ has generators $X_{ij}(a)$ for $i \neq j$ in $\{1, \ldots, n\}$ and relations given by the "universal" relations of elementary matrices. We can take the colimit $St(R)$, and it surjects onto $E(R)$. The kernel of the surjection is the center $Z(St(R))$. The corresponding short exact sequence is called the universal central extension of $E(R)$.

Definition 1.4 (1967, Milnor)**.** $K_2(R) = Z(St(R))$.

Example 1.4 (Matsumoto)**.** For a field F, $K_2(F) \cong F^\times \otimes F^\times / \langle x \otimes (1-x) \rangle$.

Remark. $K_2(R) \cong H_2(E(R), \mathbb{Z})$.

Example 1.5. $K_2(\mathbb{Z}) \cong \mathbb{Z}/2$.

Remark. $K_3(R) \cong H_3(St(R), \mathbb{Z})$. For F a field, $K_3(F) \otimes \mathbb{Q}$ is understood, by a result of Suslin.

Throughout 1968-1970, many definitions of $K_n(R)$ appeared. They are equivalent for "good" rings. Quillen's definition won out.

Definition 1.5 (Quillen)**.** Let $Pr(R)$ be the category of finitely generated projective modules. Let $QPr(R)$ be the category of finitely generated modules, with a morphism between P_1 and P_2 being a "roof diagram" (with $Q \in Pr(R)$).

$$\begin{tikzcd} & Q \arrow[dl, two heads] \arrow[dr, hook] & \\ P_1 & & P_2 \end{tikzcd}$$

We define $K_n(R) = \pi_{n+1}(B[QPr(R)])$, where $B[QPr(R)]$ is the classifying space of $QPr(R)$, i.e. a topological space built out of simplices corresponding to chains of morphism in $QPr(R)$.

Definition 1.6. For F a field, we define the Steinberg module $St_n(F)$ to have generators given by n-tuples of vectors in F^n, i.e. $[v_1, \ldots, v_n] \in F^n$. The relations are given as follows:

- If $\lambda \in F^\times$, then $[\lambda v_1, v_2, \ldots, v_n] = [v_1, \ldots, v_n]$.
- If $\sigma \in S_n$, then $[v_{\sigma(1)}, \ldots, v_{\sigma(n)}] = (-1)^\sigma [v_1, \ldots, v_n]$.
- $\sum (-1)^i [v_1, \ldots, \widehat{v_i}, \ldots, v_{n+1}] = 0$.
- If $v_1, \ldots, v_n$ are linearly dependent, then $[v_1, \ldots, v_n] = 0$.

It has a natural linear $GL_n(F)$ action.

Quillen constructed a spectral sequence $H_\bullet(GL_\bullet(F), St_\bullet(F)_\mathbb{Q}) \Rightarrow H_\bullet(B[QPr(R)]; \mathbb{Q})$.

Theorem 1.2. *$H_i(GL_n(F), St_n(F)_{\mathbb{Q}}) = 0$ for $n \geq 2$ and $i \geq n-2$.*

The group $H_{n-1}(GL_n(F), St_n(F)_{\mathbb{Q}})$ is very rich and is related to scissors congruence, polylogarithms, hyperbolic geometry...

There is another spectral sequence $H_\bullet(GL_\bullet(F), St^\infty_\bullet(F)_{\mathbb{Q}}) \Rightarrow K_0(F)_{\mathbb{Q}}$, where St^∞ is the stable Steinberg module.

2 Mar 25

We give another definition of the Steinberg module $St_n(F)$. The Tits building of F^n, J_{F^n}, is a simplicial complex where p-simplicies are partial p-flags inside F^n. A partial p-flag is a strictly increasing chain of $p+1$ nonzero proper subspaces of F^n. For $n=2$ there are only 0-simplices, which correspond to lines in F^2. For $n=3$, the 0-simplices are lines and planes in F^3, and the 1-simplices are between a line and a plane whenever the plane contains the line.

Theorem 2.1 (Solomon-Tits)**.** *J_{F^n} is homotopy equivalent to a bouquet of spheres S^{n-2}.*

The Steinberg module is $St_n(F) = \widetilde{H}_{n-2}(J_{F^n})$.

The highest dimensional chains $C_{n-2}(J_{F^n})$ are (linear combinations of) complete flags. Given a basis $v_1, \dots, v_n$ in F^n, one can associate a chain via $\sum_{\sigma \in S^n} (-1)^\sigma [\langle v_{\sigma(1)} \rangle, \langle v_{\sigma(1)}, v_{\sigma(2)} \rangle, \dots]$, and it is in fact a closed chain, hence giving an element in $St_n(F)$. This corresponds to the element $[v_1, \dots, v_n]$ from our previous definition of $St_n(F)$.

Last time we mentioned the following result:

Theorem 2.2. *$H_i(GL_n(F), St_n(F)_{\mathbb{Q}}) = 0$ for $n \geq 2$ and $i \geq n-2$.*

There is a related conjecture:

Conjecture 2.1 (Church-Farb-Pitman)**.** *$H_i(GL_n(\mathbb{Z}), St_n(\mathbb{Q})) = 0$ for $n \geq 2$ and $i \geq n-2$.*

The conjecture (should) be true for $n \leq 4$, as well as for $i = 0, 1, 2$. We will do $i = 0, 1$ later on.

Let G be a group and M a G-module. $H_0(G, M)$ is the coinvariants of M, denoted $M_G = M/(m - gm \mid m \in M, g \in G)$.

Theorem 2.3 (Ash-Rudolph)**.** *$St_n(\mathbb{Q})$ is generated as an abelian group by integral "apartments", i.e. tuples $[v_1, \dots, v_n]$ where $v_i \in \mathbb{Z}^n$ and $\det(v_1, \dots, v_n) = \pm 1$.*

Corollary 2.1. *$H_0(GL_n(\mathbb{Z}), St_n(\mathbb{Q})) = 0$ for $n \geq 2$.*

Theorem 2.4 (Bykowski)**.** *$St_n(\mathbb{Q})$ is spanned as a vector space by integral apartments subject to the following relations (for integral apartments only):*

- $[-v_1, v_2, \dots, v_n] = [v_1, \dots, v_n]$.
- *If $\sigma \in S_n$, then $[v_{\sigma(1)}, \dots, v_{\sigma(n)}] = (-1)^\sigma [v_1, \dots, v_n]$.*
- $[v_1, v_2, v_3, \dots] + [v_2, v_1 + v_2, v_3, \dots] + [v_1 + v_2, v_1, v_3, \dots] = 0$.

The last relation is similar to the relations appearing in Milnor K-theory:

Definition 2.1 (Milnor K-theory)**.** $K^M(F) = \bigoplus (F^\times)^{\otimes n}/(x \otimes (1-x))$. If one is "relaxed about 2-torsion", then $K_n^M(F) = \bigwedge^n F^\times/(x \wedge (1-x) \wedge y_3 \wedge \cdots \wedge y_n)$.

Remark. $\frac{a}{b} \wedge (1 - \frac{a}{b}) = a \wedge (b-a) - b \wedge (b-a) - a \wedge b$. (???)

2.1 End of overview

We return to discuss $K_1(R)$. Recall the elementary matrices and their subgroup $E_n(R)$ in $GL_n(R)$.

Theorem 2.5 (Suslin)**.** *$E_n(R)$ is normal in $GL_n(R)$ for R commutative and $n \geq 3$.*

Note. We assume R is commutative throughout the course even though much of the theory works with R noncommutative. However, the above theorem does require R to be commutative.

add stuff on unimodular and elementary matrices

Theorem 2.6 (Whitehead)**.** $E(R) = [GL(R), GL(R)]$.

Proof. First we show $E \subset [GL, GL]$ (we drop the R dependence notation). We use the following relation: If $i \neq k$, then $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$. Lecturer says to check it! Then $e_{ij}(a) = [e_{ik}(a), e_{kj}(1)]$ for any k different from i, j (we can pick such an index since we are working with arbitrarily large matrices). We claim that if $A, B \in GL_n$, then $ABA^{-1}B^{-1} \in E_{2n}$. We have

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -a^{-1} & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The above relation is valid even if we take $a \in GL_n$. In particular, taking $a = \mathrm{id}_n$, we get $\begin{pmatrix} 0 & -\mathrm{id}_n \\ \mathrm{id}_n & 0 \end{pmatrix} \in E_{2n}$. It follows that $\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} \in E_{2n}$. Then

$$\begin{pmatrix} A & 0 \\ 0 & A^{-1} \end{pmatrix} \begin{pmatrix} B & 0 \\ 0 & B^{-1} \end{pmatrix} \begin{pmatrix} (BA)^{-1} & 0 \\ 0 & BA \end{pmatrix} = \begin{pmatrix} ABA^{-1}B^{-1} & 0 \\ 0 & \mathrm{id}_n \end{pmatrix} \in E_{2n}.$$

We have $K_1(R) = GL(R)/E(R) = GL(R)/[GL(R), GL(R)] = H_1(GL(R), \mathbb{Z})$. We have $K_1(F) = F^\times$ and $K_1(\mathbb{Z}) = \mathbb{Z}^\times = \{\pm 1\}$.

Recall the Steinberg group defined last time. $St_n(R)$ comes with a surjection to $E_n(R)$, inducing a surjection from $\phi : St(R) \to E(R)$.

Theorem 2.7. $\ker(\phi) = Z(St(R))$.

Proof. ϕ must send $Z(St_n(R))$ into $Z(E_n(R))$. Any matrix in $Z(E_n(R))$ is diagonal, with all diagonal entries equal to the same element of $R^\times$. Upon passing to the direct limit, we see that the only element of $Z(E(R))$ is the identity. Thus $Z(St(R)) \subset \ker(\phi)$. finish proof

3 Mar 30

Let ϕ be the natural map $St(R) \to E(R)$. Let $A, B \in E(R)$ commute. Then for $a \in \phi^{-1}(A)$ and $b \in \phi^{-1}(B)$, the element $[a, b]$ depends only on A and B, and furthermore lands in $\ker(\phi) = K_2(R)$. We denote it by $A * B$.

Properties of $*$:

- $B * A = (A * B)^{-1}$.
- $(A_1 A_2) * B = (A_1 * B)(A_2 * B)$.
- $(PAP^{-1}) * (PBP^{-1}) = A * B$.

For $u, v \in R^\times$, we define

$$\{u, v\} = \begin{pmatrix} u & 0 & 0 \\ 0 & u^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix} * \begin{pmatrix} v & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & v^{-1} \end{pmatrix}.$$

fill in.
For $u, v, w \in R^\times$, we have $\{uv, w\} = \{u, w\}\{v, w\}$ and $\{u, vw\} = \{u, v\}\{v, w\}$.

Theorem 3.1. *Let $u, 1-u \in R^\times$. Then $\{u, 1-u\} = 1$ in $K_2(R)$.*

Proof. We fix $n \geq 3$ and drop R from our notation. Throughout the proof, the letters u and v will always refer to units in R.

We begin by introducing the following elements of St_n, for $1 \leq i, j \leq n$ and $i \neq j$:

$$w_{ij}(u) = x_{ij}(u) x_{ji}(-u^{-1}) x_{ij}(u).$$

As a matrix it looks like fill in. The $w_{ij}(u)$ generate a group that we call W_n. The image $\phi(W_n)$ is the monomial matrices. Now, we let $h_{ij}(u) = w_{ij}(u) w_{ij}(-1)$. As a matrix it looks like fill in. The $h_{ij}(u)$ generate a group H_n.

Exercise: $w_{ij}(u)^{-1} = w_{ij}(-u) = w_{ji}(u^{-1})$.

Lemma 3.1 (Key Lemma)**.** *Conjugation by an element of W_n sends an element of the form $x_{ij}(a)$ to an element of the form $x_{kl}(b)$, and it sends something of the form $w_{ij}(u)$ to an element of the form $w_{k'l'}(v)$.*

Let $C_n = W_n \cap \ker(\phi)$.

Corollary 3.1. *$C_n \subset Z(St_n)$.*

Proof of Corollary. Let $w \in C_n$. Then we have $\phi(w x_{ij}(a) w^{-1}) = \phi(x_{ij}(a))$. From Lemma 3.1, we can write $w x_{ij}(a) w^{-1} = x_{kl}(b)$, so we get $e_{kl}(b) = e_{ij}(a)$. Thus $i = k, j = l, a = b$, so $w x_{ij}(a) w^{-1} = x_{ij}(a)$. Thus $w \in Z(St_n)$.

Lemma 3.2. $[h_{12}(u), h_{13}(v)] = h_{13}(uv)h_{13}(u)^{-1}h_{13}(v)^{-1}$.

Proof of Lemma 3.2. todo

Note that $\{u, v\} = [h_{12}(u), h_{13}(v)]$, since the matrices on the right-hand side represent our definition of $\{u, v\}$.

Finally, let u, v be units with $u + v = 1$. By Lemma 3.2, we want to show $h_{13}(uv) = h_{13}(u)h_{13}(v)$. Symbolically it is enough to show $h_{12}(uv) = h_{12}(u)h_{12}(v)$. This is equivalent to $w_{12}(uv) = w_{12}(u)w_{12}(-1)w_{12}(v)$. By the exercise, we can write the RHS as $w_{12}(u)w_{21}(1)w_{12}(v)$, which by definition is $w_{12}(u)x_{21}(1)x_{12}(-1)x_{21}(1)w_{12}(v)$. finish proof.

Remark. Lemma 3.1 is very powerful and essentially encodes all of the Steinberg relations we will use. We can use it to compute conjugation in the Steinberg group by computing them in matrices.

Remark. Recall Milnor K_2 is $K_2^M(F) = F^\times \otimes F^\times/(u \otimes (1-u))$. The theorem implies there is a map $K_2^M(F) \to K_2(F)$. It is a theorem of Matsumoto that this map is an isomorphism. A similar result holds for local rings.

We now do some work with $K_2^M(F)$. We use additive notation for this group, and we use $\{x, y\}$ to denote the image of $x \otimes y$ in $K_2^M(F)$. Some observations:

- $\{x, -x\} = 0$. todo
- $\{x, y\} = -\{y, x\}$. todo Thus $2\{x, x\} = 0$.

Theorem 3.2. $K_2^M(\mathbb{F}_q) = 0$.

Proof. Let x generate $\mathbb{F}_q^\times$. We show that $\{x, x\} = 0$. If q is even, this follows from $\{x, -x\} = 0$. If q is odd, todo

Definition 3.1. A map from $K_2^M(F)$ to an abelian group A is called a **symbol**.

To give a symbol $c : K_2^M(F) \to A$, it is equivalent to assign elements $c(x, y) \in A$ for all units $x, y \in F$, such that $c(xy, z) = c(x, z) + c(y, z)$ and $c(x, 1-x) = 0$.

Example 3.1. There is a symbol $K_2^M(\mathbb{R}) \to \mathbb{Z}/2$ given by $c(x, y) \neq 0$ if and only if $x, y < 0$.

Example 3.2 (Tame symbol)**.** Let F be a field with a discrete valuation ν. Let $\mathcal{O}_F$ be the valuation ring, $\mathfrak{m}_F$ its maximal ideal, and $\overline{F}$ the residue field. Let $\pi \in F^\times$ have $\nu(\pi) = 1$. Then there is a map $\mathrm{res} : K_2^M(F) \to \overline{F}^\times$ satisfying

$$\mathrm{res}\{x, y\} = (-1)^{\nu(x)\nu(y)}\overline{y^{\nu(x)}x^{-\nu(y)}},$$

where the overline means image in $\overline{F}$. In terms of π and units $u, v \in \mathcal{O}_F^\times$, we have

$$\mathrm{res}\{\pi^n u, \pi^m v\} = (-1)^{nm}\overline{v^n u^{-m}}.$$

This map is characterized by the fact that $\mathrm{res}\{u, v\} = 0$ and $\mathrm{res}\{\pi, u\} = \overline{u}$.

Exercise: show $\mathrm{res}\{x, 1-x\} = 0$.

Theorem 3.3 (Tate)**.** *There is a short exact sequence*

$$0 \to \mathbb{Z}/2 \to K_2^M(\mathbb{Q}) \xrightarrow{\bigoplus_p \mathrm{res}_p} \bigoplus_p \mathbb{F}_p^\times \to 0.$$

The appearance of $\mathbb{Z}/2$ *is due to* $K_2(\mathbb{Z}) = \mathbb{Z}/2$.